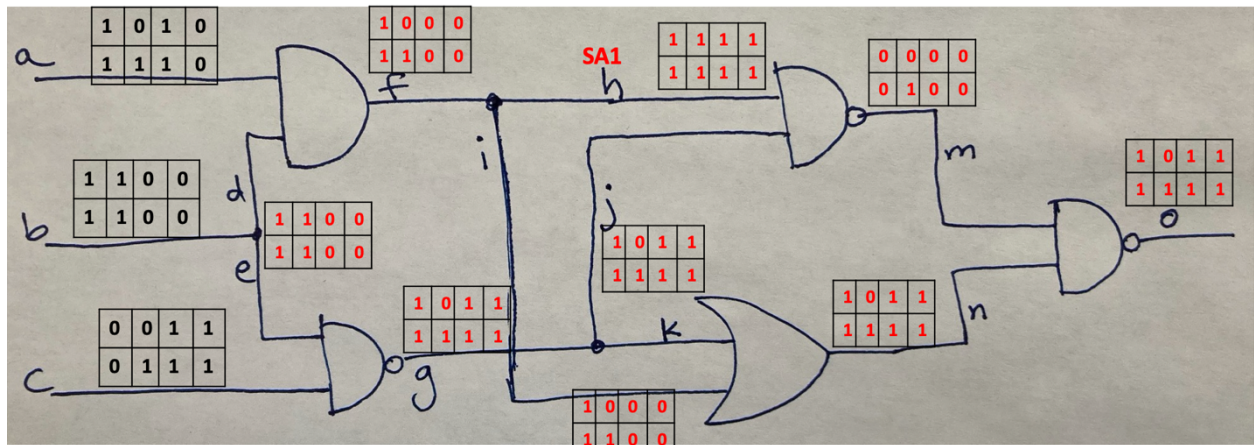


Homework # 4

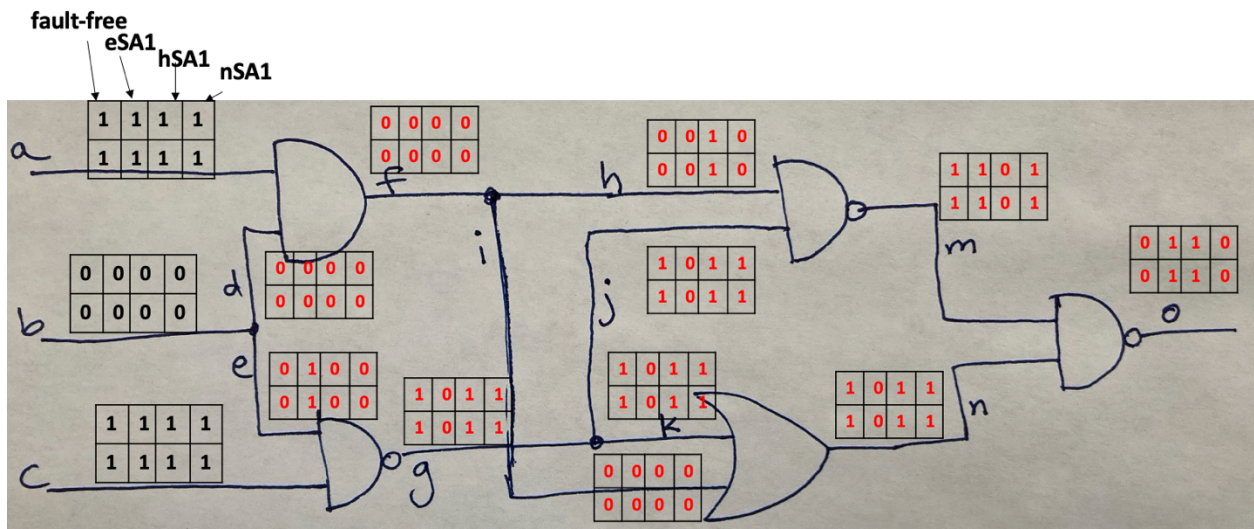
Solution

- For the circuit shown below, perform Parallel Pattern Single Fault Simulation targeting the fault h stuck-at-1. You are given 4 vectors to simulate and the vectors are encoded as 11 for 1, 00 for 0 and 01 for U.

Note: The two output bits of each gate should be calculated to ensure the above encoding of '1', '0', and 'U' are honored.

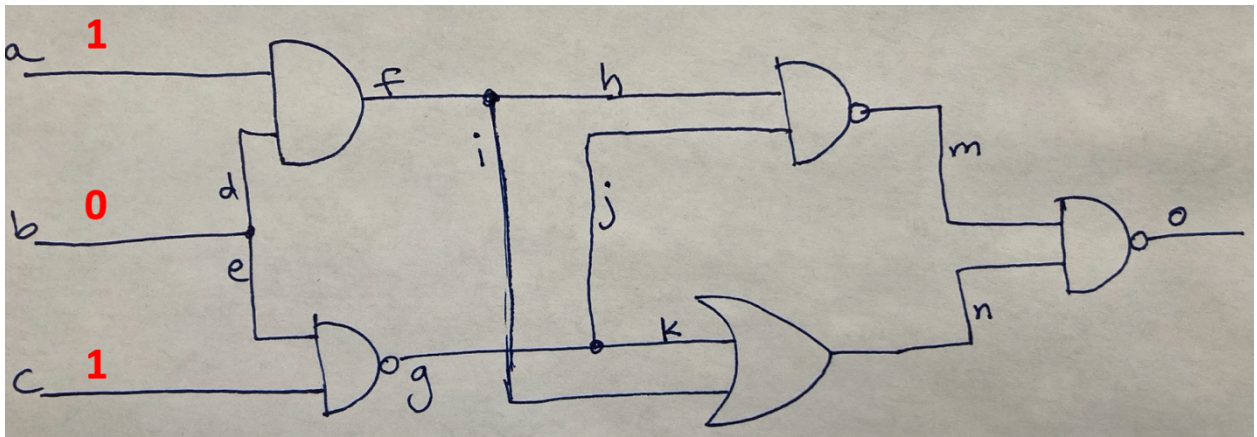


- For the circuit shown below, which is the same circuit as problem 1 but with different data, perform Parallel Fault simulation for the vector abc = 101 and three faults: e stuck-at-1, h stuck-at-1 and n stuck-at-1 to determine which faults will be detected at the primary output.



The following faults will be detected at PO : e SA1, h SA1 but n SA1 cannot be detected.

3. For the circuit of Problem 1, perform deductive fault simulation to determine all faults that will be detected at the primary output 'o' for the test vector $a=1$ $b=0$ $c=1$.



Please note, while the solution below uses a formula for each gate (using intersection and union operations), you would only need to use the definition to solve this problem.

$$L_a = \{a/0\}$$

$$L_b = \{b/1\}$$

$$L_c = \{c/0\}$$

$$L_d = \{b/1, d/1\}$$

$$L_e = \{b/1, e/1\}$$

$$L_f = L_d \cap \overline{L_a} \cup L_f = \{b/1, d/1, f/1\}$$

$$L_g = L_e \cap \overline{L_c} \cup L_g = \{b/1, e/1, g/0\}$$

$$L_h = L_f \cup L_h = \{b/1, d/1, f/1, h/1\}$$

$$L_i = L_f \cup L_i = \{b/1, d/1, f/1, i/1\}$$

$$L_j = L_g \cup L_j = \{b/1, e/1, g/0, j/0\}$$

$$L_k = L_g \cup L_k = \{b/1, e/1, g/0, k/0\}$$

$$L_m = L_h \cap \overline{L_j} \cup L_m = \{d/1, f/1, h/1, m/0\}$$

$$L_n = L_k \cap \overline{L_i} \cup L_n = \{e/1, g/0, k/0, n/0\}$$

$$L_o = L_m \cup L_n \cup L_o = \{d/1, e/1, f/1, g/0, h/1, k/0, m/0, n/0, o/1\}$$